This assignment will not be graded and is only for practice.

1) If $A + 4I = 0$, where A is a 3×3 matrix and I is an identity matrix of order 3, then find out the $det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -64

1 point

2) If $\{v_1, v_2, v_3\}$ forms a basis of R^3, then which of the following are true? ***2 points***

- ☐ $\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3.
- ☐ $\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3.
- ☐ $\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3.
- ☐ $\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3.
$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3.
$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3.
$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3.

3) Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of R^2. Choose the set of correct options. ***2 points***

- ☐ $\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3.
- ☐ $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3.
- ☐ $\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of R^3.
- ☐

$\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of R^3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3.
$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3.

4) Suppose $A = \begin{bmatrix} -3 & 5 & -1 \\ 0 & -2 & 3 \\ 7 & -4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 & x \\ x & -5 & 2 \\ -2 & 7 & 0 \end{bmatrix}$, where $x \in \mathrm{N}$. If $det(A) = det(B)$, then find the value of x.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

2 points

5) Consider the following system of linear equations:

$$x - 3y + 2z = 5 - a$$
$$3x - 4y + z = a + 1$$
$$-3x + 9y - 6z = 2a$$

where $a \in \mathrm{R}$. If the given system of linear equations has infinitely many solutions, then what is the value of a?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 15

2 points

6) Consider the set $S = \{(1, -1, 0), (0, 1, 1), (2, 0, 1)\}$. Choose the correct set of options. ***2 points***

- ☐ $(7, -2, 2) \in \text{Span}(S)$
- ☐ $\text{Span}(S)$ is a proper subspace of R^3.
- ☐ $\text{Span}(S)$ =Span$\{(1, -1, 0), (0, 1, 1)\}$

☐ dim(Span(S))=3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, -2, 2) \in$ Span(S)
dim(Span(S))=3.

7) Suppose V is a vector space defined as $V = \{A \mid A \in M_{3\times3}(\mathrm{R})$, the sum of the diagonal entries of A is zero $\}$. What is the dimension of V ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

2 points

8) Consider the following sets with the given operations defined as follows: ***2 points***

$V_1 = \{(a, b) \mid a, b \in \mathrm{R}\}$
Addition: $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2)$;
$(a_1, b_1), (a_2, b_2) \in V$

Scalar multiplication: $\alpha(a, b) = (\alpha a, b)$;
$(a, b) \in V, \alpha \in \mathrm{R}$

$V_2 = \{(a, b, c) \mid a, b, c \in \mathrm{R}$ and $ab + bc = 0\}$
Addition: $(a_1, b_1, c_1) + (a_2, b_2, c_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2)$;
$(a_1, b_1, c_1), (a_2, b_2, c_2) \in V$

Scalar multiplication: $\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$;
$(a, b, c) \in V, \alpha \in \mathrm{R}$

Choose the set of correct options.

☐ V_1 is a vector space with respect the given operations.

☐ V_1 is not a vector space with respect the given operations.

☐ V_2 is a vector space with respect the given operations.

- [ ] V_2 is not a vector space with respect the given operations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_1 is not a vector space with respect the given operations.
V_2 is not a vector space with respect the given operations.

9) Find out the correct statements from the following: ***2 points***

- [ ] The set of all invertible matrices of order 2 forms a subspace of $M_2(\mathrm{R})$ with respect to usual addition and scalar multiplication.
- [ ] The vector space R^5(with respect to usual additional and scalar multiplication) contains two disjoint subspaces.
- [ ] There exist three linearly dependent vectors in R^3 such that none of the three is a multiple of another.
- [ ] Union of two linearly independent sets need not be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exist three linearly dependent vectors in R^3 such that none of the three is a multiple of another.
Union of two linearly independent sets need not be linearly independent.

10) If S_1 is a maximal linear independent set and S_2 is a minimal spanning set of a vector space V, then which of the following option(s) is (are) true? ***2 points***

- [ ] For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
- [ ] For any $v \in S_2, S_2 \setminus \{v\}$ is a spanning set of V.
- [ ] For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
- [ ] For any $v \in V, S_2 \cup \{v\}$ is a spanning set of V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
For any $v \in V, S_2 \cup \{v\}$ is a spanning set of V.

11) Consider the following statements: ***1 point***

Statement 1: $0(0,0) + 0(1,1) = (0,0)$ implies that the set $\{(0,0),(1,1)\}$ is a linearly independent subset of R^2.
Statement 2: $2(1,0) + 2(0,1) = (2,2)$ implies that the set $\{(1,0),(0,1),(2,2)\}$ is a linearly dependent subset of R^2.

Which of the following options is correct?

- ◯ Both the statements are true.
- ◯ Statement 1 is true, but Statement 2 is false.
- ◯ Statement 2 is true, but Statement 1 is false.
- ◯ Both the statements are false.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Statement 2 is true, but Statement 1 is false.

Three people denoted by P_1, P_2, P_3 intend to buy some rolls, buns and cakes from a shop. They pay ₹ x_1, ₹ x_2, ₹ x_3 per unit for rolls, buns and cakes, respectively. P_1 bought 1 unit of rolls, 2 units of buns, and 3 units of cakes, P_2 bought 2 units of rolls, 4 units of buns, and 5 units of cakes and P_3 bought 1 unit of rolls, 3 units of buns, and 4 units of cakes. The total amount spent by P_1, P_2, P_3 are ₹ 16, ₹ 29, ₹ 21, respectively. Answer the following questions using this given data.

12)

Represent the system of linear equations above in terms of its matrix form $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 16 \\ 29 \\ 21 \end{bmatrix}$. Find the $(2, 2)$-th cofactor of A?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

13) If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is the solution of $Ax = b$, then the value of $x_1 + x_2 + x_3$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

2 points

Let $(M_{m\times n}(\mathrm{R}))$ denote the set of all $(m \times n)$ matrices with real entries and let (D) be a non-zero matrix such that (D) is the product of the matrices $(A \in M_{7\times 12}(\mathrm{R}), B \in M_{12\times 3}(\mathrm{R}))$ and $(C \in M_{3\times 5}(\mathrm{R}))$.

14) Find the maximum possible rank of (D)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

15) Find the minimum possible rank of (D)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/25